COMPUTER VISION

Comprehensive Study Notes

B.Tech Level | Complete Module Coverage

Unit I: Image Formation, Processing & Enhancement

Unit II: Depth Estimation & Multi-Camera Views

Topics: Image Formation • Convolution • Filtering • Enhancement Depth Estimation • Fundamental Matrix • Epipolar Geometry Homography • Rectification • DLT • RANSAC • Camera Calibration Histogram Processing • Feature Descriptors • Stereo Vision

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UNIT I – IMAGE FORMATION & PROCESSING

1. Fundamentals of Image Formation

Definition

Image formation is the process of capturing a 3D real-world scene and mapping it onto a 2D digital image plane through a camera or optical system.

Key Concepts

- **Image:** A 2D array of pixel values representing intensity/colour of a scene.
- **Pixel:** Smallest unit of a digital image; stores colour/intensity value.
- **Resolution:** Total number of pixels (width × height).
- **Colour Models:** RGB (Red-Green-Blue), Grayscale, HSV, YCbCr.
- **Image point:** p = (u, v) in 2D image coordinates.
- **3D world point:** P = (X, Y, Z) in camera coordinate system.

Pinhole Camera Model

The pinhole camera model is the simplest model relating 3D world points to 2D image points. Light passes through a tiny pinhole and forms an inverted image.

$$x = f \cdot X / Z \quad y = f \cdot Y / Z$$

Where: f = focal length, Z = depth, (X,Y,Z) = 3D point, (x,y) = 2D image point

```
3D World Point P(X,Y,Z)
|
| (light ray)
|
Image Plane [===] Pinhole O -----> Z-axis
at z = f |
|
p(x,y) on image

Similar triangles: x/f = X/Z => x = fX/Z
```

Difference: Image vs. CV vs. Text Processing

Aspect	Image Processing	Computer Vision	Text Processing
Input	Raw pixel data	Images/video	Text/characters
Goal	Enhance/filter images	Understand scene/objects	Analyse language
Output	Improved image	Decisions/descriptions	Structured data
Example	Blurring, sharpening	Object detection	Sentiment analysis

Techniques	Convolution, filters	ML, geometry	NLP, regex

2. Convolution and Filtering

Definition

Convolution is a mathematical operation that combines an input image with a kernel (filter matrix) to produce a transformed output image. It is the foundation of image filtering in Computer Vision.

Convolution Formula

$$(I * K)[i,j] = \Sigma \Sigma I[i+m, j+n] \cdot K[m, n]$$

*I = Input image, K = Kernel/Filter, * = Convolution operator*

Convolution Steps

- Flip the kernel horizontally and vertically (180°).
- Slide the flipped kernel over each pixel of the image.
- Multiply overlapping values element-wise.
- Sum all products → output pixel value.
- Repeat for every pixel position.

Worked Example – 3×3 Convolution

```
Image patch (3x3): Kernel (3x3):
| 1 2 3 | | 0 0 0 |
| 4 5 6 | | 0 1 0 |
| 7 8 9 | | 0 0 0 |

This is the identity kernel → output = 5 (centre pixel preserved)

Gaussian Blur Kernel (3x3): Sharpen Kernel (3x3):
| 1 2 1 | / 16 | 0 -1 0 |
| 2 4 2 | | -1 5 -1 |
| 1 2 1 | | 0 -1 0 |
```

Common Filters and Their Effects

Filter	Kernel / Method	Effect
Mean (Box) Filter	All 1s / (n^2)	Blurs image, removes noise
Gaussian Filter	Bell-curve weights	Smooth blur, natural look
Sobel Filter	Edge-detect kernels	Detects horizontal/vertical edges
Laplacian Filter	4-neighbour diff	Detects all edges (2nd derivative)
Sharpen Kernel	Centre-5 kernel	Enhances edges/detail
Median Filter	Non-linear, sort	Removes salt-and-pepper noise

Key Properties of Convolution

- **Commutative:** f * g = g * f
- **Associative:** (f * g) * h = f * (g * h)

- **Distributive:** f * (g + h) = f*g + f*h
- **Linearity:** Convolution is a linear operation.
- **Padding:** Zero-padding preserves image size after convolution.

■ *In practice, cross-correlation (no kernel flip) is often used and called "convolution" in deep learning frameworks.*

3. Image Enhancement

Definition

Image enhancement is the process of improving the visual quality or accentuating certain features of an image to make it more suitable for a specific application. It does not add new information — it manipulates what is already present.

Categories of Enhancement

- **Spatial Domain:** Direct pixel manipulation (point, neighbourhood operations).
- **Frequency Domain:** Transform image (e.g., Fourier), then filter, then inverse.

Spatial Domain Techniques

Technique	Formula / Description	Use
Contrast Stretching	`I' = (I − min)/(max − min) × 255`	Expand dynamic range
Gamma Correction	`I' = c · I^γ (γ<1 brightens)`	Monitor calibration
Log Transform	`I' = c · log(1 + I)`	Compress bright regions
Negative Image	`I' = L − 1 − I`	Enhance dark detail
Thresholding	`I' = 255 if I ≥ T, else 0`	Binary segmentation
Histogram Eq.	`Redistribute pixel intensities`	Improve contrast

Frequency Domain Enhancement

The image is converted to frequency domain using the Fourier Transform. Filters are applied in frequency domain, then converted back.

- **Low-pass filter:** Passes low frequencies → smoothing/blurring.
- **High-pass filter:** Passes high frequencies → edge sharpening.
- **Band-pass filter:** Passes specific frequency range.

$$F(u,v) = \Sigma \Sigma f(x,y) \cdot e^{(-j2\pi(ux/M + vy/N))} \text{ [2D DFT]}$$

4. Histogram Processing

Definition

A histogram of an image represents the frequency (count) of each intensity value in the image. It provides global information about the tonal distribution.

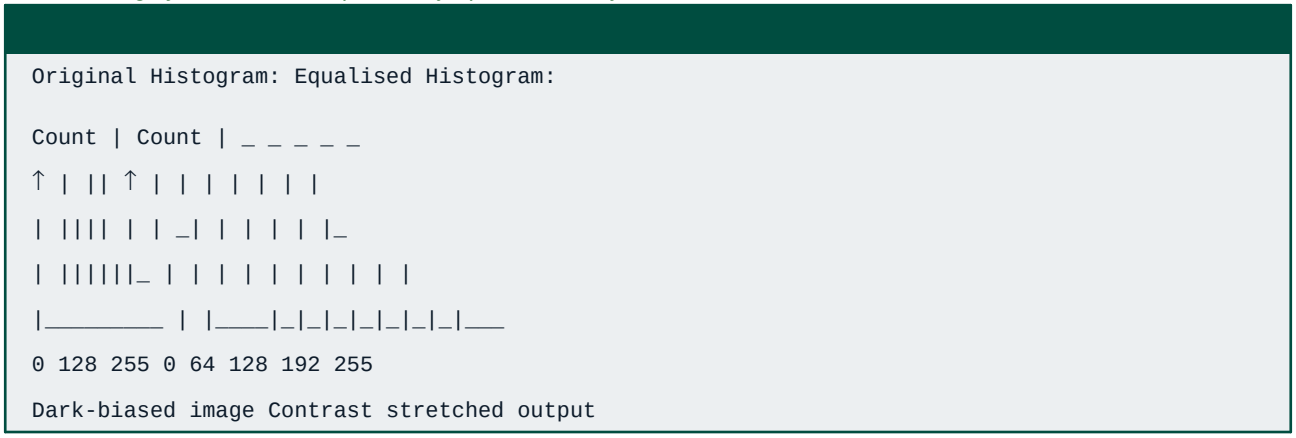
h(r_k) = n_k where r_k = intensity level, n_k = number of pixels

Histogram Equalization (HE)

Histogram equalization redistributes pixel intensities to achieve a uniform distribution, maximising the image contrast.

$s_k = T(r_k) = (L–1) \cdot \Sigma (n_j / n)$ for j = 0 to k

L = total grey levels, n = total pixels, n_j = pixels at level j



Steps for Histogram Equalization

- **1.** Compute histogram h(r_k) for all intensity levels.
- **2.** Compute normalised histogram: p(r_k) = h(r_k) / (M × N).
- **3.** Compute cumulative distribution function (CDF).
- **4.** Map original intensities to new values using the CDF formula.
- **5.** Apply mapping to produce the equalised image.

Histogram Specification (Matching)

Histogram specification transforms an image so that its histogram matches a desired (target) histogram — more control than simple equalisation.

- Equalise source image to get intermediate s.
- Equalise target histogram to get intermediate v.
- Find inverse mapping from v to target intensities.
- Compose mappings to transform source → target histogram.

Applications

- Medical imaging: enhance X-ray or MRI contrast.
- Satellite imagery: improve land feature visibility.
- Fingerprint enhancement for criminal identification.
- Pre-processing for machine learning pipelines.

5. Feature Descriptors in Image Processing

What is a Feature?

A feature is a distinctive or informative part of an image used to identify, match, or classify objects. Good features are repeatable, distinctive, and invariant to transformations.

Types of Features

- **Edges:** Sharp boundaries between regions of different intensity.
- **Corners:** Points with high curvature or intensity change in all directions.
- **Blobs:** Regions with consistent appearance differing from surroundings.
- **Ridges:** Elongated structures like roads or vessels.
- **Texture:** Repetitive patterns in a region.

Harris Corner Detector

Detects corners by computing the second-moment matrix M of image gradients.

$$R = \det(M) - k \cdot \text{trace}(M)^2 \quad k \approx 0.04–0.06$$

- R >> 0: corner (feature point).
- R << 0: edge.
- R ≈ 0: flat region.

SIFT – Scale-Invariant Feature Transform

SIFT detects and describes local features invariant to scale, rotation, and partial illumination changes.

- **1.** Scale-space extrema detection using Difference of Gaussians (DoG).
- **2.** Keypoint localisation – reject low-contrast/edge points.
- **3.** Orientation assignment – dominant gradient direction.
- **4.** Keypoint descriptor – 128-D vector of gradient histograms.

Other Important Descriptors

Descriptor	Key Property	Dimensions
SIFT	Scale + rotation invariant	128-D
SURF	Faster than SIFT (Hessian)	64-D
ORB	Free, fast (binary)	256-bit
HOG	Shape/gradient histogram	Variable
LBP	Texture, local binary	Variable
BRIEF	Binary, very fast	128/256-bit

6. Projection – Perspective & Parallel

3D Projection – Definition

Projection is the mapping of 3D world points onto a 2D view plane. The projection line is called a Projector. The point where all projectors converge is the Projection Reference Point (PRP) or Centre of Projection.

Types of Projection

```
Projection
/ \
Parallel Perspective
/ \ |
Orthographic Oblique (1-pt, 2-pt, 3-pt)
/ \ / \
Top Front Cavalier Cabinet
Side Axonometric
```

Parallel Projection

Projection lines are parallel to each other. The Centre of Projection is at infinity. The image preserves the actual size of the object.

- **Orthographic:** Projectors perpendicular to view plane. Subtypes: Top, Front, Side views.
- **Axonometric:** Rotated coordinate system. Shows 3 faces (Isometric most common).
- **Oblique:** Projectors at an angle. Subtypes: Cavalier (true length), Cabinet (half length).

Perspective Projection

Projection lines converge at the Centre of Projection (eye point). Objects appear smaller with increasing distance — mimics human vision.

$$x' = x / (1 - z/z_c) \quad y' = y / (1 - z/z_c)$$

Where z_c = distance from origin to Centre of Projection

Types of Perspective Projection

- **1-Point (One Vanishing Point):** View plane parallel to one principal axis. Parallel lines meet at 1 VP. Example: railway track viewed head-on.
- **2-Point (Two Vanishing Points):** View plane intersects two principal axes. Corner-view of buildings.
- **3-Point (Three Vanishing Points):** View plane not parallel to any principal axis. Extreme wide-angle/aerial views.

Vanishing Point

Parallel lines in 3D that are not parallel to the view plane appear to converge at a point on the horizon called the Vanishing Point.

Perspective Projection Matrix (1-Point)

$$M = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/z_c & 1 \end{vmatrix}$$

■ *Centre of projection: (0, 0, z_c). Vanishing point: (0, 0, –z_c). The value r = –1/z_c is the key parameter in the perspective matrix.*

7. Homography

Definition

A homography is a projective transformation (3×3 matrix H) that maps points from one image plane to another. It is valid for planar scenes or pure camera rotation.

p' ~ H · p (equality up to a scale factor)

Homogeneous Coordinates

Homogeneous coordinates represent a 2D point (x, y) as a 3-vector (x, y, 1).

[x'] [h11 h12 h13] [x] [y'] = [h21 h22 h23] [y] [w'] [h31 h32 h33] [1] Convert back: X' = x'/w' Y' = y'/w'

Properties of Homography Matrix H

- 3×3 matrix with 8 independent parameters (1 scale degree of freedom).
- Minimum 4 non-collinear point correspondences required to compute H.
- Works exactly for planar scenes or camera rotation only.
- Encodes all projective transformations: scale, rotation, shear, perspective.

Applications of Homography

Application	Description
Image Stitching	Aligns overlapping images for panorama creation using common keypoints.
Document Scanning	Converts a tilted photo of a document into a flat rectangular view.
Perspective Correction	Removes projective distortion from planar surfaces.
AR Marker Tracking	Tracks flat markers in video and overlays virtual content.
Bird's-Eye View	Transforms road camera view into top-down view (Inverse Perspective Mapping).
Camera Calibration	Relates image planes from different viewpoints of a calibration board.

8. Fourier Transform in Image Processing

Definition

The Fourier Transform decomposes an image into its constituent spatial frequencies. Low frequencies represent slow intensity changes; high frequencies represent edges and fine details.

$$F(u,v) = \sum_x \sum_y f(x,y) \cdot e^{-j2\pi(ux/M + vy/N)}$$

$$f(x,y) = (1/MN) \sum_u \sum_v F(u,v) \cdot e^{j2\pi(ux/M + vy/N)} \text{ [Inverse DFT]}$$

- **Magnitude Spectrum:** $|F(u,v)|$ — shows which frequencies are present.
- **Phase Spectrum:** $\angle F(u,v)$ — encodes spatial arrangement.
- **DC Component:** $F(0,0)$ = average intensity (brightest point at centre after shift).
- **Low frequencies:** Centred at origin → global structure, background.
- **High frequencies:** Away from origin → edges, noise, fine detail.

■ *Convolution in spatial domain ↔ Multiplication in frequency domain. This is the Convolution Theorem: f * g = F■¹{F(f) · F(g)}*

9. Applications – Criminal Detection, Medical & Defense

Criminal Detection

- **Face Recognition:** Detects and matches face features (eigenfaces, deep CNNs). CCTV footage → suspect database matching.
- **Fingerprint Recognition:** Extracts minutiae (ridge endings, bifurcations). Loops → whorls → arches classification.
- **Iris Recognition:** Gabor wavelet features of iris texture. Unique per person.
- **Gait Analysis:** Identifies individuals by walking pattern from video.
- **Vehicle Plate Recognition (ANPR):** OCR on licence plates from camera feeds.

Medical CV Applications

- **X-ray / CT / MRI Analysis:** Tumour detection, organ segmentation.
- **Drug Discovery:** Microscopy image analysis of cell structures and drug effects.
- **Retinal Scanning:** Detects diabetic retinopathy from fundus images.
- **Histopathology:** Classifies tissue samples using deep learning.
- **Surgical Robotics:** Real-time guidance using CV for precise incisions.

Defense Sector

- **Object Detection:** Identify vehicles, weapons from satellite/drone imagery.
- **Autonomous Navigation:** Unmanned vehicles using depth + stereo CV.
- **Target Tracking:** Multi-object tracking in surveillance video.
- **Camouflage Detection:** IR + visible image fusion to locate hidden objects.

UNIT II – DEPTH ESTIMATION & MULTI-CAMERA VIEWS

10. Depth Estimation – Introduction

Definition

Depth Estimation is the process of measuring the distance of objects in a scene from the camera. Images are 2D (u, v), but the real world is 3D (X, Y, Z). Depth estimation recovers the missing Z dimension.

Image point: p = (u, v) 3D point: P = (X, Y, Z) RGB tells WHAT is visible; Depth tells WHERE it is in 3D

Why Depth is Ambiguous in a Single Image

The pinhole projection equations x = fX/Z and y = fY/Z give only 2 equations for 3 unknowns (X, Y, Z). Therefore, depth Z cannot be uniquely determined from a single image — this is the fundamental ambiguity.

- **Size-Distance Ambiguity:** Small nearby object looks identical to large distant object. Image size ∝ (actual size / depth).
- **Scale Ambiguity:** Only relative depths are known; absolute metric scale unknown.
- **Occlusion Ambiguity:** Objects hidden behind others cannot be measured.

Depth Estimation Techniques

Method	Principle	Key Note
Stereo Vision	Disparity between 2 cameras	`Z = fB/d`
Monocular (DL)	CNN learns depth cues	`Single image → depth map`
LiDAR	Laser pulse time-of-flight	`Very accurate, expensive`
Time-of-Flight Camera	IR light travel time	`Real-time, short range`
Structured Light	Known pattern deformation	`Used in Kinect, Face ID`
Motion-Based	Optical flow from video	`Near = faster apparent motion`

Depth Map

A depth map is an image where each pixel value encodes the distance of that point from the camera. It has the same spatial dimensions as the RGB image.

- Bright pixels → close to camera (small Z).
- Dark pixels → far from camera (large Z).
- Output size = same as input image.

11. Binocular (Stereo) Vision

Definition

Binocular stereo vision uses two cameras (mimicking human eyes) placed side by side to estimate depth through the principle of triangulation. The horizontal offset of the same point in the two images is called disparity, which is inversely proportional to depth.



Key Symbols

Symbol	Meaning
C_L, C_R	Centres of left and right cameras
B	Baseline: distance between the two cameras
P(X,Y,Z)	3D point in the scene
x_L, x_R	Horizontal pixel positions of P in left/right images
d = x_L − x_R	Disparity (horizontal shift)
f	Focal length (same for both cameras)
Z	Depth of point P from camera baseline

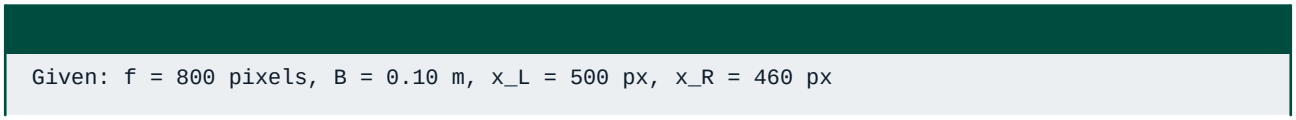
Derivation of Stereo Depth Formula

Using the pinhole camera model for both cameras:

Left camera: $x_L = f \cdot X / Z \rightarrow X = Z \cdot x_L / f$ Right camera: $x_R = f \cdot (X - B) / Z = x_L - fB/Z$
Disparity: $d = x_L - x_R = fB / Z$ Therefore: $Z = f \cdot B / d$

■ *Large disparity d → object is NEAR (small Z). Small disparity d → object is FAR (large Z).*

Worked Numerical Example



```
Step 1: d = x_L − x_R = 500 − 460 = 40 pixels

Step 2: Z = fB / d = (800 × 0.10) / 40 = 80 / 40 = 2 m

Answer: The point is 2 metres away from the camera.
```

Stereo Correspondence Problem

Finding matching points between left and right images is the hardest step. Challenges include textureless surfaces, repeated patterns, occlusion, reflections, and low light.

Depth error: $\Delta Z \approx -(fB / d^2) \cdot \Delta d$ [error grows rapidly for small disparities]

From Disparity Map to Depth Map

- Compute disparity d(u,v) for every pixel position (u,v).
- Apply Z(u,v) = fB / d(u,v) for each pixel.
- Result: a dense depth map of the full scene.
- Combined with pixel (u,v) and camera parameters → full 3D point cloud.

3D Point Recovery from Depth Pixel

$Z = f_x \cdot B / d$ [depth from disparity] $X = (u - c_x) \cdot Z / f_x$ [horizontal world coord] $Y = (v - c_y) \cdot Z / f_y$ [vertical world coord] Full 3D point: P = (X, Y, Z)

Binocular Vision – Functional Summary

Stage	Description
Camera Setup	Two parallel cameras separated by baseline B, both with focal length f
Image Capture	Both cameras capture the same 3D scene simultaneously
Rectification	Images warped so corresponding rows are aligned horizontally
Stereo Matching	For each left-image pixel, find corresponding pixel in right image on same row
Disparity Calc	d = x_L – x_R computed for every matched pixel pair
Depth Recovery	Z = fB/d converts disparity to metric depth
Depth Map	Dense Z-values assembled into a grayscale depth map image

12. Camera Geometry & Parameters

Camera Geometry

- **Camera Centre C:** Point from which the scene is observed (optical origin).
- **Optical Axis:** Imaginary line through camera centre, perpendicular to image plane.
- **Image Plane:** Surface at distance f where 2D image is formed.
- **Principal Point (c_x, c_y):** Where optical axis meets image plane.
- **Focal Length f:** Distance from camera centre to image plane.

Intrinsic Camera Matrix K

The intrinsic matrix encodes all internal optical properties of the camera. It transforms 3D camera coordinates to 2D pixel coordinates.

[f_x s c_x] K = [0 f_y c_y] [0 0 1] f_x, f_y = focal lengths in pixels (c_x, c_y) = principal point s = skew factor

Projection Equations (Full)

u = c_x + f_x · X / Z v = c_y + f_y · Y / Z λ [u, v, 1]■ = K [X, Y, Z]■ (λ = Z, scale factor)

Extrinsic Camera Parameters

Extrinsic parameters describe the camera's position and orientation in the world coordinate system. They transform world coordinates to camera coordinates.

- **R** (3×3 Rotation matrix): Orientation of camera in world frame.
- **t** (3×1 Translation vector): Position of camera centre in world.

X_cam = R (X_world – C) → X_cam = R · X_world + t (t = –RC) Complete Projection: λ [u, v, 1]■ = K [R | t] [X_w, Y_w, Z_w, 1]■ ■■■ Intrinsic ■■■■■■ Extrinsic ■■■

Rotation Matrices (3D)

R_x(θ) = | 1 0 0 | R_y(θ) = | cos θ 0 sin θ | | 0 cos θ –sin θ | | 0 1 0 | | 0 sin θ cos θ | |–sin θ 0 cos θ | R_z(θ) = | cos θ –sin θ 0 | | sin θ cos θ 0 | | 0 0 1 |

Summary: Intrinsic vs Extrinsic

Parameter	Type	Role	Matrix Size
`f_x, f_y, c_x, c_y, s`	Intrinsic	3D camera coords → 2D pixel	3×3 (K)
`R, t (or C)`	Extrinsic	3D world coords → 3D camera	3×4 [R|t]
`K[R|t]`	Projection P	3D world coords → 2D pixel	3×4

13. Epipolar Geometry

Definition

Epipolar geometry describes the geometric relationship between two cameras observing the same 3D scene. It constrains the search for corresponding points from a 2D problem to a 1D search along a line — the epipolar line.

Key Components

- **Camera Centres C■, C■:** Optical centres of left and right cameras.
- **Baseline:** Line segment connecting C■ and C■.
- **Epipolar Plane:** Plane containing C■, C■, and the 3D point P.
- **Epipoles e■, e■:** Projections of C■ onto right image (e■) and C■ onto left image (e■). Where baseline meets image plane.
- **Epipolar Lines:** Intersection of epipolar plane with each image plane. Corresponding point must lie on its epipolar line.
- **Correspondence:** If x in left image, matching point x' lies on epipolar line l' = F·x in right image.

```
P(X,Y,Z)
/ \
C■ ---+ +--- C■
| epipolar plane |
Image 1 Image 2
[e■]---l■ l■---[e■]
| |
x■ x■ (must lie on l■)

Epipolar Constraint: x■■ F x■ = 0

Key Insight: Reduces 2D search for x■ to 1D search along epipolar line l■
```

Epipolar Constraint (Practical)

For any point x in image 1, its corresponding point x' in image 2 must lie on the epipolar line l' = F·x. This is the epipolar constraint.

x'■ F x = 0 [Epipolar Constraint]

Why Epipolar Geometry Matters

- Reduces stereo matching from 2D search to 1D search → faster and more accurate.
- Enables computation of depth from stereo without knowing 3D structure.
- Foundation for computing the Fundamental Matrix and Essential Matrix.
- After rectification, epipolar lines become horizontal → row-by-row matching.

14. Fundamental Matrix

Definition

The Fundamental Matrix F is a 3×3 matrix that encodes the epipolar geometry between two uncalibrated cameras. It relates corresponding points in two views without requiring knowledge of camera intrinsics.

x'■ F x = 0 for all corresponding pairs (x, x')

Properties of F

- 3×3 matrix, defined up to a scale factor.
- Rank 2 (singular matrix) — det(F) = 0.
- 7 degrees of freedom (9 entries – 1 scale – 1 rank constraint).
- Minimum 7 point correspondences required to compute F.
- With 8+ points, can use the 8-point algorithm (linear solution).
- Epipolar line in image 2 for point x: l' = F · x.
- Epipolar line in image 1 for point x': l = F■ · x'.
- Epipoles: F · e = 0 and F■ · e' = 0.

Computing F – The 8-Point Algorithm

- **1.** Collect ≥ 8 corresponding point pairs (x_i, x'_i).
- **2.** Normalise points: translate to centroid, scale so mean distance = √2.
- **3.** Set up linear system: Af = 0, where each row is built from one correspondence.
- **4.** Solve using SVD: f = last column of V in A = UΣV■.
- **5.** Reshape f into 3×3 matrix F■.
- **6.** Enforce rank-2: compute SVD of F■, set smallest singular value to 0, recompose.
- **7.** Denormalise: F = N'■ F■ N.

Example: Building F from Two Images

```
Two images of same scene taken at different angles:

Image 1 points (x): Image 2 points (x'):
x■ = (120, 200, 1) x'■ = (80, 198, 1)
x■ = (300, 150, 1) x'■ = (255, 148, 1)
x■ = (450, 280, 1) x'■ = (400, 277, 1)
... (≥8 pairs needed)

Constraint: x'■■ F x■ = 0 for each pair
Build matrix A (8×9), solve Af = 0 by SVD
Result F used to compute epipolar lines:
l' = F · x■ → x'■ must lie on l'
```

Essential Matrix E

When camera intrinsics K■ and K■ are known, use the Essential Matrix E instead of F. E encodes only the rotation R and translation direction t between calibrated cameras.

E = [t]× R (t is translation, R is rotation) F = K■■■ E K■■¹ (F relates to E via intrinsics) x'■ E x = 0 (calibrated epipolar constraint)

■ *E has 5 DOF (3 rotation, 2 translation direction — scale unknown). Minimum 5 point correspondences needed (5-point algorithm).*

F vs E – Comparison

Property	Fundamental Matrix F	Essential Matrix E
Calibration	Not required (uncalibrated)	Intrinsics K must be known
DOF	7	5
Min Correspondences	7 (non-linear) or 8 (linear)	5
Recovers	Epipolar geometry only	R and t (up to scale)
Relation	F = K■■■ E K■■¹	E = K■■ F K■

15. Homography Rectification

What is Rectification?

Image rectification is the process of warping two stereo images so that corresponding points lie on the same horizontal row (scanline). This simplifies the stereo matching problem to a 1D search along rows.

- Before rectification: epipolar lines may be oblique in both images.
- After rectification: epipolar lines become horizontal and row-aligned.
- Rectification is achieved by applying a homography to each image.

```
BEFORE RECTIFICATION: AFTER RECTIFICATION:

Image 1 Image 2 Image 1 Image 2

[ x■ ] [ ] [ x■ ] [ ]

[ \ / x'■ ] [■■ x■■■]■■■■■[■■ x'■]

[ epipolar lines ] [ same row matches! ]

[ oblique/angled ] [■■ x■■■]■■■■■[■■ x'■]

y_L ≠ y_R (difficult) y_L = y_R (1D search only)
```

Rectification Pipeline

- **1.** Calibrate both cameras (get K■, K■, R, t).
- **2.** Compute rectifying rotation matrices R■ and R■ for each camera.
- **3.** Apply homography H■ = K · R■ to left image.
- **4.** Apply homography H■ = K · R■ to right image.
- **5.** After warping, all epipolar lines are horizontal.
- **6.** Stereo matching reduces to row-by-row 1D search.

Homography for Rectification

The rectifying homographies rotate both camera coordinate systems so that the baseline becomes parallel to the horizontal image axis, making epipolar lines horizontal in both rectified images.

Rectified image 1: I'■ = warp(I■, H■) where H■ = K[R■ | 0] Rectified image 2: I'■ = warp(I■, H■) where H■ = K[R■ | 0] After rectification: y_L = y_R for all corresponding point pairs

■ *Rectification is essential in real stereo systems (automotive, robotics) because it reduces computation time for stereo matching dramatically.*

16. Direct Linear Transformation (DLT)

Definition

DLT is a linear algebraic method for estimating projective transformations (homography, camera projection matrix) from point correspondences. It converts the estimation problem into a system of linear equations solved by Singular Value Decomposition (SVD).

DLT for Homography – Setup

Given point x = (x, y, 1)■ and corresponding point x' = (x', y', 1)■, the relationship x' ~ H x expands to:

Each correspondence gives 2 equations: [–x –y –1 0 0 0 x'x x'y x'] h = 0 [0 0 0 –x –y –1 y'x y'y y'] h = 0 h = [h11, h12, h13, h21, h22, h23, h31, h32, h33]■ (9-vector)

DLT Algorithm Steps

- **Collect correspondences:** Need ≥ 4 non-collinear pairs for homography; ≥ 6 pairs for camera matrix P.
- **Normalise points:** Translate centroid to origin; scale so mean distance = √2. This critical step improves numerical stability.
- **Build matrix A:** Stack 2 rows per correspondence → A is (2n × 9) for homography.
- **Solve Ah = 0:** Compute SVD: A = UΣV■. Solution h = last column of V (corresponding to smallest singular value).
- **Reshape:** Reshape the 9-vector h into the 3×3 matrix H.
- **Denormalise:** H = N'■¹ H■ N where N, N' are normalisation transforms.
- **Evaluate:** Measure reprojection error ε = Σ d(x'■, H x■)².

SVD – Key Role in DLT

SVD decomposes A = UΣV■. The solution to Ah = 0 that minimises ■Ah■ subject to ■h■ = 1 is the right singular vector corresponding to the smallest singular value.

A = U Σ V■ → h = last column of V

Applications of DLT

- Computing homography matrix H from image correspondences.
- Estimating camera projection matrix P from 3D-2D correspondences.
- Computing Fundamental matrix F (used in the 8-point algorithm).
- 3D triangulation: finding 3D point from two 2D projections.

■ *DLT is fast and simple but sensitive to outliers. In practice, always combine DLT with RANSAC for robustness.*

17. RANSAC – Random Sample Consensus

Definition

RANSAC is a robust estimation algorithm that fits a model to data containing outliers (incorrect matches). It works by repeatedly sampling minimal sets of points, fitting a model, and counting how many other points agree with (are inliers to) that model.

RANSAC Algorithm

```
Input: Set of N data points (many may be outliers)
Output: Best-fit model + set of inliers

REPEAT for N_iter iterations:
1. SAMPLE: Randomly select minimal subset s (e.g., 4 for homography)
2. FIT: Estimate model from this minimal sample (e.g., compute H)
3. SCORE: Count inliers = points where error < threshold ε
error = reprojection distance d(x', H·x)
4. KEEP: If inlier count > best so far, save this model

FINAL: Refit model using ALL inliers of best model (lower error)
```

Number of Iterations Formula

N = log(1 − p) / log(1 − w^s) p = desired probability of success (e.g., 0.99) w = estimated inlier ratio (e.g., 0.5 = 50% inliers) s = minimum sample size (e.g., 4 for homography, 8 for F)

Worked Example

```
Problem: Compute homography H with 50% outliers in matches.
p = 0.99 (want 99% success), w = 0.5, s = 4 (min for H)

N = log(1 − 0.99) / log(1 − 0.5^4)
= log(0.01) / log(1 − 0.0625)
= (−4.605) / (−0.0645)
≈ 71 iterations needed

With 70% inliers (w=0.7): N ≈ 12 iterations (much fewer needed!)
```

RANSAC Threshold Selection

- **Too small ε:** Many inliers missed, model may not converge.
- **Too large ε:** Outliers included, model quality drops.
- **Typical ε:** 1–3 pixels reprojection error for homography/F.

DLT + RANSAC – Standard Pipeline

```
Raw Feature Matches (many outliers)
|
```

```
v
RANSAC: Random 4-point sample → DLT → compute H
↓ Count inliers (d(x', Hx) < ε)
Repeat N times, keep best H
|
v
Refit DLT using ALL inliers → Final H
|
v
Low reprojection error, robust result
```

RANSAC vs DLT – Comparison

Property	DLT alone	DLT + RANSAC
Handles outliers	No	Yes
Approach	Linear least squares	Iterative sampling
Complexity	O(n)	O(N_iter × s)
Accuracy	Degrades with outliers	Robust to ~50%+ outliers
Requires	≥ min points	N iterations

18. Camera Calibration & Auto-Calibration

Definition

Camera calibration is the process of estimating camera parameters (intrinsic K and extrinsic R, t) so that precise 3D-to-2D projection and 2D-to-3D back-projection can be performed.

Why Calibration is Needed

- Undistort images (lens distortion correction).
- Measure real-world distances from images.
- Enable stereo depth estimation with metric scale.
- Augmented reality: correctly overlay 3D objects on images.
- Robot vision: relate pixel position to real-world coordinates.

Zhang's Method – Chessboard Calibration

The most widely used calibration method. Uses images of a planar chessboard pattern from multiple viewpoints.

- **1.** Print a chessboard pattern (e.g., 9×6 inner corners).
- **2.** Capture 10–20 images of the board from different angles.
- **3.** Detect chessboard corner positions in each image (sub-pixel accuracy).
- **4.** For each image, set up homography H from 3D board → 2D image.
- **5.** Stack constraints from all images → solve for K.
- **6.** Use K to initialise non-linear optimisation (Levenberg-Marquardt).
- **7.** Jointly optimise K, distortion coefficients k■, k■, p■, p■, and all R, t.

Lens Distortion Model

Radial distortion: x_d = x_u(1 + k■r² + k■r■ + k■r■) Tangential: x_d = x_u + 2p■xy + p■(r² + 2x²) r² = x² + y² (distance from principal point)

- Barrel distortion (k■ < 0): edges bend outward — fisheye lenses.
- Pincushion distortion (k■ > 0): edges bend inward — telephoto lenses.

Auto-Calibration

Auto-calibration estimates intrinsic parameters K from image correspondences alone, without a known calibration target. It exploits constraints on K derived from the fundamental matrix or absolute conic.

- **Kruppa Equations:** Relate fundamental matrix F to intrinsic matrix K.
- **Absolute Conic:** A conic in 3D whose image is related to K.
- **Image of Absolute Conic (IAC):** ω = K■■ K■¹; self-consistent across views.
- **Modulus Constraint:** Uses orthogonality of rotation matrices.

■ *Auto-calibration is less accurate than target-based methods but useful when calibration patterns are not available (e.g., internet photos, legacy footage).*

Calibration Output Parameters

Parameter	Symbol	Typical Value
Focal length (horiz)	f_x	500–4000 pixels
Focal length (vert)	f_y	Similar to f_x

Principal point X	c_x	image_width / 2
Principal point Y	c_y	image_height / 2
Radial distortion 1	k■	−0.3 to +0.3
Radial distortion 2	k■	−0.1 to +0.1
Tangential distortion	p■, p■	Near 0
Rotation matrix	R	3×3 orthogonal
Translation vector	t	3×1 meters

19. 3D Reconstruction & Multi-Camera Views

3D Reconstruction Pipeline

```
IMAGES → FEATURE DETECTION → FEATURE MATCHING
|
v
FUNDAMENTAL/ESSENTIAL MATRIX ESTIMATION
|
v
CAMERA POSE ESTIMATION (R, t per image)
|
v
TRIANGULATION (3D points)
|
v
BUNDLE ADJUSTMENT (joint optimisation)
|
v
SPARSE POINT CLOUD → DENSE RECONSTRUCTION
```

Triangulation

Triangulation recovers the 3D position of a scene point from its projections in two or more calibrated images. Given x■ ~ P■ X and x■ ~ P■ X, find X.

For point x in image 1 and x' in image 2: x × (P X) = 0 → A X = 0 (DLT applies!) Build A from projection equations, solve via SVD → 3D point X

Bundle Adjustment

Bundle adjustment jointly refines all camera parameters (K, R, t) and 3D point positions to minimise the total reprojection error across all views.

min Σ■ Σ■ d(x_ij, P_i X_j)² over all camera matrices P_i and 3D points X_j

■ *Solved by Levenberg-Marquardt (LM) algorithm. This is the final polishing step in all structure-from-motion and SLAM systems.*

Multi-Camera View Considerations

- **Stereo:** 2 cameras, fixed baseline → metric depth from disparity.
- **Trinocular:** 3 cameras → better matching, handles occlusion better.
- **Wide-Baseline Stereo:** Cameras far apart → better depth accuracy for distant objects.
- **Omnidirectional Stereo:** 360° coverage using multiple cameras on a rig.
- **Camera Networks:** Multiple cameras with overlapping fields of view for surveillance.

SLAM – Simultaneous Localisation and Mapping

SLAM uses moving cameras to simultaneously build a 3D map of the environment while tracking camera pose. Combines visual odometry, loop closure, and bundle adjustment.

- Visual Odometry: frame-to-frame camera motion estimation.
- Loop Closure: recognise previously-visited places to correct drift.
- Dense Mapping: build detailed 3D models using depth cameras or stereo.
- Applications: autonomous vehicles, drones, AR headsets, robots.

QUICK REFERENCE – KEY FORMULAS & FACTS

Stereo Depth	Z = fB/d | d = x_L − x_R | Large d → near; Small d → far
Pinhole Projection	x = f·X/Z | y = f·Y/Z | λ[u,v,1]■ = K[X,Y,Z]■
Homography	p' ~ Hp | H is 3×3, 8 DOF | ≥4 non-collinear pts needed
Fundamental Matrix	x'■Fx = 0 | Rank 2, 7 DOF | ≥8 pts (linear), ≥7 pts (non-linear)
Essential Matrix	E = [t]×R | F = K■■■ E K■■¹ | 5 DOF, ≥5 pts
DLT	Solve Ah = 0 | SVD: h = last col of V | Normalise first!
RANSAC	N = log(1−p)/log(1−w^s) | Score by inlier count | Refit on all inliers
Intrinsic K	K = [fx, s, cx; 0, fy, cy; 0, 0, 1] | Intrinsic: 3D cam → 2D px | Extrinsic [R|t]: world → cam
Convolution	(I*K)[i,j] = ΣΣ I[i+m,j+n]·K[m,n] | Flip kernel, slide, multiply, sum | Spatial domain filter
Histogram Eq.	h(r_k) = n_k | s_k = (L−1)·CDF(r_k) | Maximises contrast